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Towards a unified molecular mechanism for ligand-dependent activation of NR4A-RXR heterodimers

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eLife Assessment

This **important** study investigated whether the nuclear receptor Nur77 is regulated by a non-canonical mechanism of ligand-induced disruption of its interaction with RXR γ , similar to the family member Nurr1. The overall evidence is **compelling**. This manuscript will be of interest to scientists focusing on mechanisms of transcriptional regulation.

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Abstract

A subset of nuclear receptors (NRs) function as permissive heterodimers with retinoid X receptor (RXR), defined by transcriptional activation in response to RXR agonist ligands. Permissive NR-RXR activation is generally understood to operate through a classical pharmacological mechanism in which RXR agonist binding enhances coactivator recruitment to the heterodimer. However, we previously demonstrated that transcriptional activation of permissive Nurr1-RXR α (NR4A2-NR2B1) heterodimers by an RXR ligand set, which included pharmacological RXR agonists and selective Nurr1-RXR α agonists that function as antagonists of RXR α homodimers, is explained by a non-classical activation mechanism involving ligand-binding domain (LBD) heterodimer dissociation (Yu et al., 2023). Here, we extend mechanistic ligand profiling of the same RXR ligand set to the evolutionarily related Nur77-RXR γ (NR4A1-NR2B3) heterodimer. Biochemical and NMR protein-protein interaction profiling, together with cellular transcription studies, indicate that activation of Nur77-RXR γ transcription by the RXR ligand set, which lacks selective Nur77-RXR γ agonists, is consistent with contributions from both classical pharmacological activation and LBD heterodimer dissociation. However, reanalysis of our previously published data for Nurr1-RXR α revealed that inclusion of selective Nurr1-RXR α agonists was essential for elucidating the LBD heterodimer dissociation mechanism. Together, our findings highlight the importance of using a more functionally diverse RXR ligand set to define the mechanism of Nur77-RXR γ activation and to further evaluate whether LBD heterodimer dissociation represents a shared activation mechanism among NR4A-RXR heterodimers relevant to neurodegenerative and inflammatory diseases.

Introduction

The orphan nuclear receptor Nur77 (NR4A1) is implicated as a drug target in central nervous system (CNS) and neurological disorders including stroke and Parkinson's disease (Liu et al., 2021 ; Saucedo-Cardenas and Conneely, 1996 ) and chronic inflammatory and autoimmune disorders (Lith and de Vries, 2021 ). Nur77 plays important roles in maintaining the health,

development, and homeostasis of the brain by regulating autophagy (Ping et al., 2021 [↗](#)), neuroinflammation (Yan et al., 2020 [↗](#)), and dopamine turnover and transmission (Lévesque and Rouillard, 2007 [↗](#)). Nur77-deficient mice exhibit higher levels of dopamine metabolite DOPAC and reduced levels of dopamine alongside increased spontaneous locomotor activity (Gilbert et al., 2006 [↗](#); Mount et al., 2013 [↗](#)). Nur77 is protective against Parkinsonian symptoms induced by neurotoxin 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) treatment (Mount et al., 2013 [↗](#)), emphasizing the relevance of Nur77 to neurodegenerative disease. Nur77 activates the NLRP3 inflammasome through a non-transcriptional regulation mechanism through direct binding to NLRP3 (Zhu et al., 2023 [↗](#)) and anti-tumor immunity via transcriptional regulation (Liebmann et al., 2018 [↗](#); Mandula et al., 2024 [↗](#)). Nur77 also regulates oxidative metabolism in skeletal muscle tissues and cells (Chao et al., 2012 [↗](#), 2007 [↗](#)) and plays an important role in thymic negative selection in T-cells (Liebmann et al., 2018 [↗](#); Lith et al., 2020 [↗](#)).

The molecular basis for the druggability of NRs is primarily encoded within a specific region, the ligand-binding domain (LBD), which has evolved to bind endogenous cellular metabolites and, in turn is the target for approximately 15% of FDA-approved drugs (Santos et al., 2017 [↗](#)). The orthosteric ligand-binding pocket is a solvent-occluded hydrophobic region located in the core of the LBD. Agonist binding to the orthosteric pocket stabilizes the NR LBD in a conformation that enables high affinity binding and recruitment of transcriptional coactivator proteins to increase transcription and gene expression (Kojetin and Burris, 2013 [↗](#))—the classical pharmacological activation mechanism—but can also impact nuclear import/export and subcellular localization, post-translational modification, interactions with other proteins, among other activities. However, crystal structures of the LBDs of Nur77 and an evolutionarily related NR4A subfamily member, Nurr1 (NR4A2), show that their orthosteric ligand-binding pockets are filled with bulky hydrophobic residues (Flaig et al., 2005 [↗](#); Wang et al., 2003 [↗](#)), which may prevent ligand access. Progress has been made to identify putative Nur77 and Nurr1 endogenous cellular metabolites (Bruning et al., 2019 [↗](#); de Vera et al., 2019 [↗](#), 2016; Rajan et al., 2020 [↗](#); Vinayavekhin and Saghatelian, 2011 [↗](#)) and synthetic ligands (Jang et al., 2021 [↗](#); Munoz-Tello et al., 2020 [↗](#); Safe et al., 2021 [↗](#); Willems and Merk, 2022 [↗](#)) that bind to their LBDs and regulate their activities. However, since the discovery of Nurr1 and Nur77 in the late 1980s and early 1990s (Hazel et al., 1988 [↗](#); Law et al., 1992 [↗](#)), small molecule targeting the NR4A receptors remains challenging compared to other prototypical NRs (Burris et al., 2013 [↗](#)), leading the field to seek out alternative mechanisms to regulate NR4A activities.

Nurr1 and Nur77 can regulate transcription as monomers or as a permissive heterodimer with retinoic X receptors RXR α (NR2B1) and RXR γ (NR2B3), respectively, which form heterodimers with approximately one-third of the NR superfamily (Lefebvre et al., 2010 [↗](#); Lévesque and Rouillard, 2007 [↗](#)). Thus, one such alternative mechanism to target NR4A transcription that has been explored in the field is through ligands binding to their RXR heterodimer partner (Asvos et al., 2025 [↗](#); Loppi et al., 2018 [↗](#); McFarland et al., 2013 [↗](#); Spathis et al., 2017 [↗](#); Wallen-Mackenzie et al., 2003 [↗](#); Wang et al., 2016 [↗](#)). NR-RXR heterodimers are classified as permissive or non-permissive depending on whether they can be activated by RXR agonist binding to RXR (permissive) or not (non-permissive). Pharmacological activation of NRs is understood to occur via a classical model where agonist binding stabilizes an active LBD conformation that promotes recruitment of transcriptional coactivator proteins, ultimately resulting in increased gene expression. However, we recently showed that transcriptional activation of Nurr1-RXR α heterodimers by pharmacological RXR agonists or selective Nurr1-RXR α agonists, which function as antagonists of RXR α homodimers, occurs through a non-classical model: LBD heterodimer protein-protein interaction (PPI) inhibition (Yu et al., 2023 [↗](#)).

Building on our published Nurr1-RXR α study, we hypothesized that Nur77 activity could similarly be modulated by ligand binding to its heterodimer partner, RXR γ . Here, we tested the same RXR ligand set used in our Nurr1-RXR α study using the same comparative Nur77-RXR γ structure-function analyses. Together, our published and new findings show that RXR γ and RXR α function as repressive heterodimer partners for Nur77 and Nurr1, respectively. Ligand profiling studies also indicate that RXR ligands may influence Nur77-RXR γ heterodimer transcription through LBD

heterodimer dissociation. However, nuanced observations in the Nur77-RXRy ligand profiling data—and the lack of selective Nur77-RXRy agonists that, similar to selective Nurr1-RXRα agonists, would function as antagonists of RXR homodimers—limit the ability to determine the extent to which Nur77-RXRy activation reflects contributions from LBD heterodimer dissociation, classical pharmacological activation, or both mechanisms.

Results

RXRy LBD is required for repression of Nur77-mediated transcription

In our recently study (Yu et al., 2023 [DOI](#)) we confirmed published observations (Aarnisalo et al., 2002 [DOI](#); Forman et al., 1995 [DOI](#)) that RXRα represses Nurr1-mediated transcription on a monomeric Nurr1-binding DNA response element called NBRE, which is present in the promoter regions of Nurr1-regulated genes in the dopamine biosynthesis pathway. Using RXRα domain truncation analysis, we previously found that RXRα-induced repression occurs via the RXRα LBD interaction with Nurr1, consistent with the well-understood mechanism that LBD-LBD interactions constitute the primary basis for NR homodimerization and heterodimerization. We also showed that RXR ligands that activate transcription of Nurr1-RXRα heterodimers function via a mechanism involving LBD-LBD heterodimer PPI inhibition, freeing a transcriptionally active Nurr1 monomer from the repressive Nurr1-RXRα heterodimer. This was in contrast to the prevailing mechanism in the field for RXR ligand-induced activation of permissive heterodimers where, for example, RXRα contributes to activation of PPARy-mediated transcription (Kliewer et al., 1992 [DOI](#)), and agonist binding to either or both NRs synergistically activates PPARy-RXRα heterodimer-mediated transcription via enhanced coactivator recruitment (de Vera et al., 2017 [DOI](#); DiRenzo et al., 1997 [DOI](#); Kojetin et al., 2015 [DOI](#)).

To test if RXRy represses Nur77-mediated transcription through a similar mechanism with Nurr1-RXRα, we performed a transcriptional reporter assay where SK-N-BE(2) neuronal cells were transfected with a full-length Nur77 expression plasmid, with or without full-length or domain-truncation RXRy expression plasmids (Figure 1a [DOI](#)), along with a plasmid containing three copies of the monomeric NBRE DNA-binding response element sequence upstream of luciferase gene (3xNBRE-luc). Similar to our Nurr1-RXRα findings (Yu et al., 2023 [DOI](#)), cotransfection of RXRy (FL) repressed Nur77-mediated transcription, whereas a RXRy construct lacking the LBD (ΔLBD) had no effect (Figure 1b [DOI](#)). Cotransfection of an RXRy construct lacking the N-terminal AF-1 domain (ΔNTD) activated Nur77-mediated transcription, whereas a construct lacking the NTD and DNA-binding domain containing the hinge region (RXRy-LBD) had no effect.

These data, which indicate that both the RXRy NTD and LBD are necessary for repression of Nur77-mediated transcription, are consistent with a model in which LBD-dependent PPIs contribute to RXR-mediated repression of Nur77-RXRy and Nurr1-RXRα heterodimers. However, there are additional nuances that should be considered in the interpretation of these data. First, the RXRy NTD is a principal domain that drives RXRy phase separation *in vitro* and biomolecular condensates in cells (Sotys et al., 2025 [DOI](#), 2021 [DOI](#); Wang et al., 2025 [DOI](#)). Therefore, perturbing RXRy's ability to form cellular condensates via removal of its NTD may change how RXRy interacts with Nur77, which also forms cellular condensates (Chen et al., 2025 [DOI](#), 2024 [DOI](#); Peng et al., 2021 [DOI](#)), or coregulator proteins that are recruited to condensates containing Nur77 and/or RXRy. Furthermore, the NTD of nuclear receptors can interact with coactivator and corepressor proteins (Dotzlaw et al., 2002 [DOI](#); Hodgson et al., 2005 [DOI](#); Simons et al., 2014 [DOI](#); Wang and Simons, 2005 [DOI](#)), including a study that showed the Nur77 NTD interacts with the FHL2 corepressor (Kurakula et al., 2011 [DOI](#)). Although we are not aware of published studies showing the RXRy NTD interacts with corepressor proteins, in principle truncation of the RXRy NTD could show higher transcription via decreased corepressor interaction to Nur77-RXRy(ΔNTD) heterodimers relative to wild-type Nur77-RXRy heterodimers. Together, these and other limitations of the RXRy domain truncation strategy—including potential differences in protein expression levels and truncation-dependent changes in subcellular localization, which we did not assess—limit a full mechanistic interpretation of

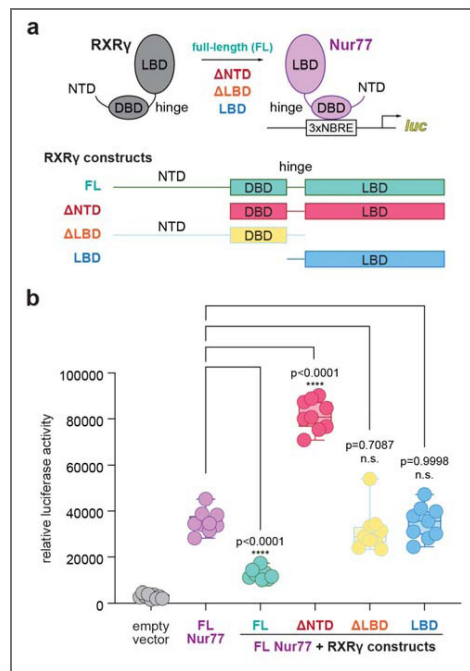


Figure 1. Contribution of RXRγ domains on repressing Nur77-mediated transcription.

(a) General scheme of the cellular transcriptional reporter assay. **(b)** 3xNBRE-luciferase assay performed in SK-N-BE(2)-C cells. Data are normalized to empty vector control (n=9 replicates), shown as a box and whiskers plot with boundaries of the box representing the 25th percentile and the 75th percentile, and representative of three independent experiments. Statistical testing was performed and p-values were calculated using the Brown-Forsythe and Welch multiple comparisons test of the FL Nur77 + RXRγ constructs conditions relative to FL Nur77 control condition. See **Figure 1—source data 1** for data plotted.

these data. Despite these limitations, the truncation data support a conclusion that both the RXR γ NTD and LBD contribute to repression of Nur77-mediated transcription and thus point to a more complex domain-level mechanism of Nur77-RXR γ regulation than observed for Nurr1-RXR α .

RXR ligand profiling of Nur77-RXR γ heterodimer activation vs. pharmacological RXR γ agonism

We next determined the influence of RXR ligands on Nur77-RXR γ transcriptional activity using the 3xNBRE luciferase reporter assay in human SK-N-BE(2) neuronal cells (Figure 2a). We used the same RXR ligand set (Figure 2—figure supplement 1) from our previous study on Nurr1-RXR α (Yu et al., 2023) composed of pharmacological RXR agonists (9-cis retinoic acid, Bexarotene, CD3254, IRX4204, LG100268) and antagonists (Danthron, HX531, PA452, Rhein, UVI3003), two compounds reported to selectively activate Nurr1-RXR α heterodimers (BRF110 and HX600), and one compound reported to selectively activate PPAR-RXR and RAR-RXR heterodimers (LG100754). Similar to our findings for Nurr1-RXR α , pharmacological RXR agonists activated Nur77-RXR γ transcription—however, one of the RXR agonists, IRX4204, activated Nur77-RXR γ but not Nurr1-RXR α (Yu et al., 2023), which may be explained by its 2.5-fold potency for RXR γ compared to RXR α ; compound 2 in (Vuligonda et al., 2001). Pharmacological RXR antagonists did not affect Nur77-RXR γ transcription, consistent with our Nurr1-RXR α findings. For two selective Nurr1-RXR α agonists, BRF110 and HX600, which function as RXR α homodimer antagonists, HX600 showed partial activation of Nur77-RXR γ whereas BRF110 did not activate, consistent with the original report of BRF110 as a selective agonist for Nurr1-RXR α (Spathis et al., 2017).

We next profiled how the RXR ligands affect transcriptional activation of RXR γ homodimers using biochemical and cellular assays similar to the assays we performed for RXR α (Yu et al., 2023). We used a time-resolved fluorescence resonance energy transfer (TR-FRET) biochemical assay to determine how the ligands influence the interaction between RXR γ LBD and a peptide derived from PGC1 α (Figure 2b), a coactivator that regulates RXR-mediated transcription (Delerive et al., 2002). We also determined how the ligands influence transcription mediated by RXR γ homodimers (Figure 2c). Similar to our findings for RXR α , the RXR agonists enhanced PGC1 α recruitment to the RXR γ LBD and increased RXR γ transcriptional activity, whereas antagonists had no effect. BRF110 did not affect RXR γ transcriptional activity; however, HX600 activated RXR γ -mediated transcription, consistent with a previous study showing that HX600 displays high activity for RXR γ but moderate activity for RXR α (Umemiya et al., 1997). However, HX600 did not promote recruitment of the PGC1 α coactivator peptide in the TR-FRET assay, indicating that the activation of RXR γ transcription by HX600 may occur through either a coactivator recruitment-independent mechanism or through recruiting other coactivators among other possibilities, which would need further study.

In our previous study on Nurr1-RXR α , using Pearson (linear) and Spearman (monotonic relationship) analysis we found that Nurr1-RXR α mediated transcription by the RXR ligand set is not correlated to features of pharmacological RXR α agonism—i.e., RXR α homodimer transcription or RXR α LBD coactivator peptide recruitment in a TR-FRET biochemical assay—but instead show moderate-to-strong correlations to weakening Nurr1-RXR α LBD interaction and binding affinity in the NMR and ITC data, respectively (Yu et al., 2023). In contrast, a strong correlation was observed between RXR α homodimer transcription and RXR α LBD TR-FRET coactivator peptide recruitment, consistent with the classical pharmacological activation mechanism. Similarly, we observed a strong correlation is observed between RXR γ transcription and RXR γ LBD TR-FRET coactivator peptide recruitment mediated by the RXR ligand set (Figure 2d), indicating RXR α and RXR γ share a similar classical pharmacological activation mechanism. However, in contrast to what we reported for Nurr1-RXR α , we found that Nur77-RXR γ transcription imparted by the RXR ligand set is strongly correlated to with coactivator peptide recruitment to the RXR γ LBD in a TR-FRET biochemical assay (Figure 2e) and transcriptional activation of full-length RXR γ (Figure 2f). These data suggest that RXR ligands may regulate Nur77-RXR γ transcription through the classical pharmacological activation mechanism, which is distinct from Nurr1-RXR α , leading us to test if these RXR ligands can dissociate Nur77-RXR γ as they do to Nurr1-RXR α (Yu et al., 2023).

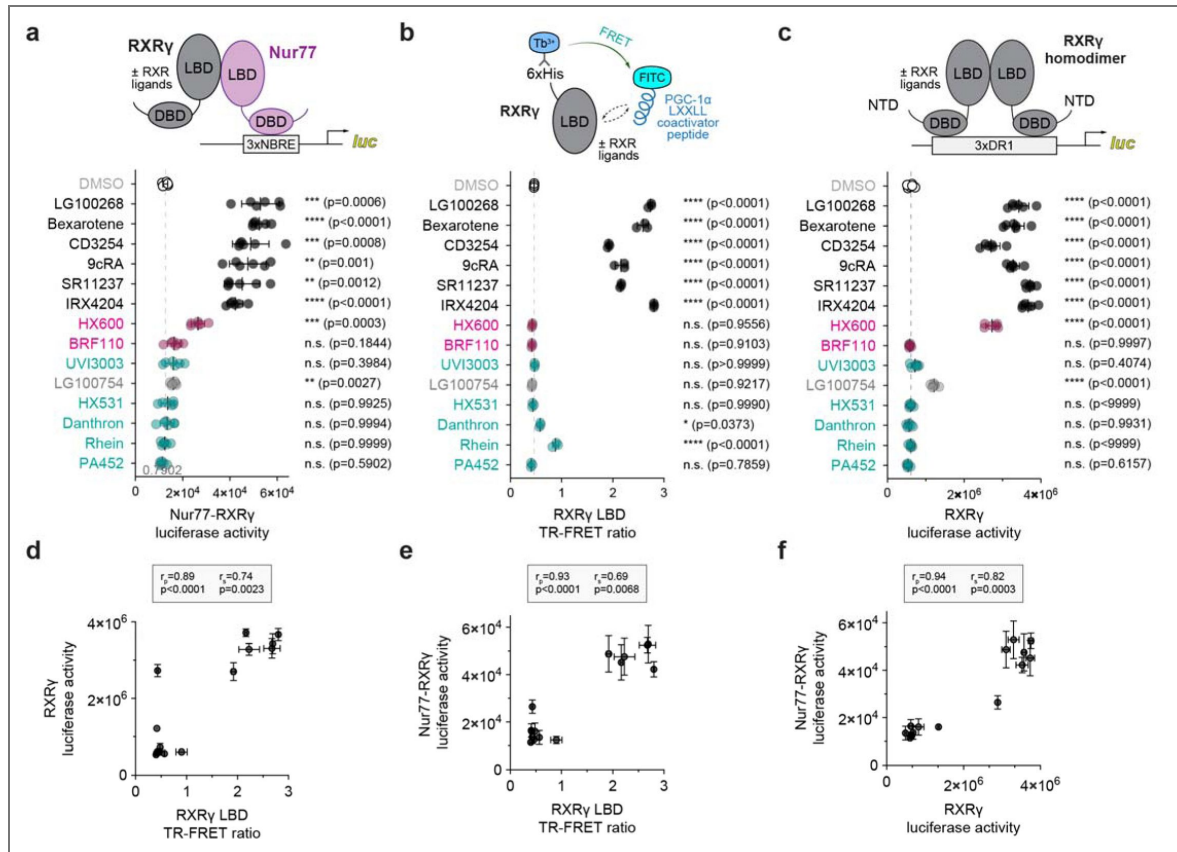


Figure 2. Ligand profiling for Nur77-RXRy heterodimer activation and pharmacological RXRy agonism.

(a) General scheme and data from the Nur77-RXRy/3xNBRE-luciferase cellular transcriptional reporter assay performed in SK-N-BE(2)-C cells treated with RXR ligand (1 μ M) or DMSO (dotted line). Data are normalized to DMSO (n=6 replicates), represent the mean $\pm$ s.d., and representative of two independent experiments. See **Figure 2—source data 1** for data plotted. (b) General scheme and data from RXRy LBD time-resolved fluorescence resonance energy transfer (TR-FRET) coactivator peptide interaction assay. TR-FRET ratio measured in the presence of DMSO (dotted line) or compound (2–4 μ M). Data are normalized to DMSO control (n=3 replicates), represent the mean $\pm$ s.d., representative of two independent experiments. See **Figure 2—source data 2** for data plotted. (c) General scheme and data from the RXRy/3xDR1-luciferase cellular transcriptional reporter assay performed in HEK293T cells treated with compound (1 μ M) or DMSO control (dotted line). Data normalized to DMSO (n=6 replicates), represent the mean $\pm$ s.d., and representative of two independent experiments. See **Figure 2—source data 3** for data plotted. (d,e,f) Correlation plots of (d) RXRy transcriptional reporter data vs. RXRy LBD TR-FRET data, (e) Nur77-RXRy cellular transcription data vs. RXRy LBD TR-FRET data, and (f) Nur77-RXRy cellular transcription data vs. RXRy transcriptional reporter data with calculated Pearson (r_p) and Spearman (r_s) correlation coefficients. For all comparisons, statistical testing was performed, and p-values were calculated, using the Brown-Forsythe and Welch (a,c) or ordinary one-way ANOVA (b) tests for multiple comparisons with Dunnett corrections relative to DMSO control treated condition. Data and RXR ligand label text in (a,b,c) are colored according to RXR ligand activity as grouped in **Figure 2—figure supplement 1**.

RXR ligand profiling of Nur77-RXR γ heterodimer activation vs. LBD heterodimer dissociation

In our previous study, the RXR ligand-induced Nurr1-RXR α LBD heterodimer dissociation mechanism was apparent via three structural biology and biophysical methods (Yu et al., 2023 [DOI](#)). 2D [^1H , ^{15}N]-TROSY-HSQC NMR structural footprinting showed that addition of Nurr1-RXR α activating ligands to ^{15}N -labeled Nurr1 LBD heterodimerized with RXR α LBD resulted in the appearance of a population of Nurr1 LBD monomer NMR peaks where monomer population occupancy strongly correlated to RXR ligand-induced Nurr1-RXR α transcriptional activation. We therefore performed NMR structural footprinting to determine the impact of the RXR ligand set on Nur77-RXR γ LBD heterodimer interaction.

We first performed multi-angle light scattering (MALS) at a protein concentration (180 μM) similar to our NMR samples (200–400 μM), which confirmed the Nur77 LBD and Nurr1 LBD are monomeric in solution, whereas RXR α and RXR γ LBDs are predominately homodimers with a minor homotetramer species (Figure 3—figure supplement 1 [DOI](#)) consistent with published studies on RXR (Gampe et al., 2000 [DOI](#); Kersten et al., 1997 [DOI](#)). Because there are no published NMR chemical shift assignments for Nur77 LBD, we expressed and purified [^2H , ^{15}N , ^{13}C]-labeled Nur77 LBD, collected three-dimensional (3D) NMR backbone assignment experiments, and assigned ~85% of non-proline chemical shifts. NMR chemical shift footprinting analysis showed that addition of RXR γ LBD to ^{15}N -labeled Nur77 LBD revealed a transition in 2D [^1H , ^{15}N]-TROSY-HSQC NMR data from a Nur77 LBD monomer species to a Nur77-RXR α LBD heterodimeric form (Figure 3—figure supplement 2 [DOI](#)). Addition of RXR ligands resulted in select chemical shift perturbations (CSPs) where a population of NMR peaks corresponding to both the Nur77 LBD monomer and the Nur77-RXR γ heterodimer were observed. This peak doubling phenomenon was apparent for several Nur77 residues with well-resolved NMR peaks, including G544 (Figure 3a [DOI](#)) and G376 (Figure 3—figure supplement 3 [DOI](#)) located on helix 9 near the heterodimerization surface and helix 1 distal from the heterodimerization surface, respectively (Figure 3—figure supplement 4 [DOI](#)).

In our previous study on Nurr1-RXR α , addition of RXR ligands resulted in the appearance of monomeric Nurr1 NMR peaks that were in slow exchange on the NMR time scale, where the monomeric and heterodimeric species were long-lived, allowing an estimation of the relative population of dissociated monomeric Nurr1 via peak integration analysis (Yu et al., 2023 [DOI](#)). However, NMR data of RXR ligands added to ^{15}N -labeled Nur77 heterodimerized to RXR γ LBD are more complex, showing mixed features of slow exchange, intermediate-to-fast exchange, and multiple populations and peak lineshapes that includes peaks shifted off the diagonal between Nur77 in the monomer and heterodimer states—and is also evident in the NMR spectrum of the heterodimer complex in the absence of ligand. At two molar equivalents of added RXR γ LBD, a shift to the heterodimer species (Figure 3a [DOI](#), green arrow) is evident with a small population in exchange with the Nur77 monomer present shifted in a different direction from the heterodimer peak (Figure 3a [DOI](#), blue arrow). These data suggest that RXR γ LBD binding to Nur77 LBD may occur through a binding exchange mechanism that includes an induced fit or a conformational change after binding component. Notwithstanding these challenges, we estimated the relative population of dissociated monomeric Nur77 LBD using measured NMR peak volumes before and after addition of RXR ligand (Figure 3b [DOI](#)), which revealed that all the RXR ligands dissociated the Nur77-RXR γ LBD heterodimer to some degree although it is important to note that this analysis is confounded by the mix of exchange and peak lineshape properties—this system does not reveal isolated monomer vs. heterodimer populations as we observed for Nurr1-RXR α . Pearson (linear) and Spearman (monotonic relationship) analysis reveals a strong correlation between the RXR ligand-induced appearance of NMR-detected Nur77 monomer populations and Nur77-RXR γ transcription activation, which indicates RXR ligands activating Nur77-RXR γ transcription may also promote heterodimer dissociation (Figure 3c [DOI](#)).

In our previous study on Nurr1-RXR α , isothermal titration calorimetry (ITC) studies showed a strong correlation between ITC-measured reduction in Nurr1-RXR α LBD heterodimerization affinity and transcriptional activation of Nurr1-RXR α for the RXR ligand set (Yu et al., 2023 [DOI](#)). We

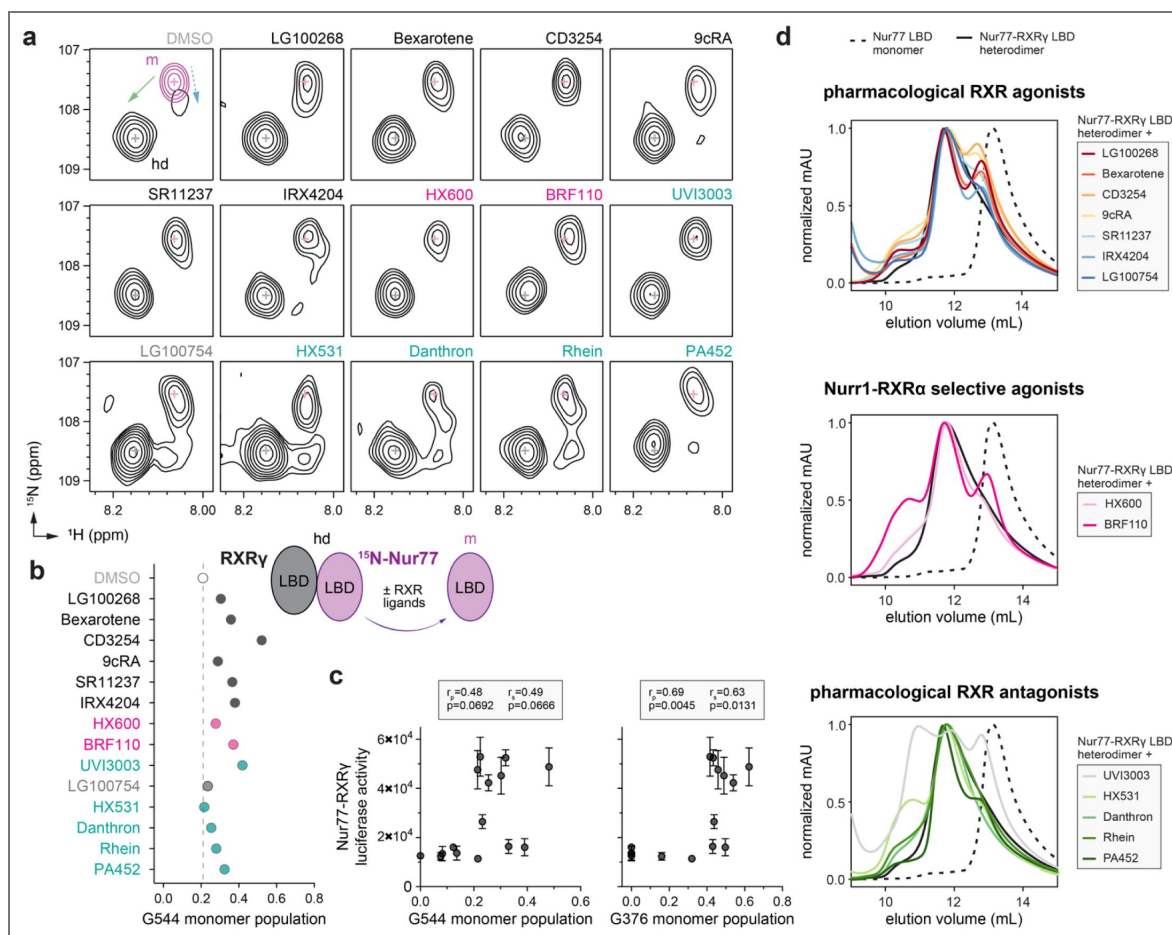


Figure 3. Ligand profiling for Nur77-RXRγ LBD heterodimer dissociation.

(a) 2D ^1H , ^{15}N -TROSY HSQC data of ^{15}N -labeled Nur77 LBD heterodimerized with unlabeled RXRγ LBD in the presence of RXR ligands focused on the NMR peak of G544. The upper left shows an overlay of two spectra corresponding to ^{15}N -labeled Nur77 LBD monomer (200 μM ; purple) and ^{15}N -labeled Nur77 LBD + unlabeled RXRγ LBD heterodimer (1:2 molar ratio; black) to demonstrate the shift of the G544 peak between Nur77 LBD monomer (m) and heterodimer (hd) forms; solid green and dotted blue arrows denote the complex chemical shift perturbation pattern. RXR ligand label text is colored according to RXR ligand activity as grouped in Figure 2—figure supplement 1. (b) Peak intensity estimated ligand-dependent Nur77 LBD monomer populations from G544 and G376 in the 2D ^1H , ^{15}N -TROSY HSQC data ($n=1$). Data and RXR ligand label text are colored according to RXR ligand activity as grouped in Figure 2—figure supplement 1. See Figure 3—source data 1 for data plotted. (c) Correlation plot of Nur77-RXRγ cellular transcription data vs. NMR estimated Nur77 LBD monomer populations for G544 and G376 with calculated Pearson (r_p) and Spearman (r_s) correlation coefficients; see Figure 3—figure supplement 3 for G376 2D ^1H , ^{15}N -TROSY HSQC data. (d) Analytical size exclusion chromatography (SEC) analysis of Nur77-RXRγ LBD in the presence of RXR ligands (solid colored lines) relative to Nur77 LBD monomer (dotted black line) and Nur77-RXRγ LBD heterodimer (solid black line) ($n=1$).

therefore used ITC to quantify how RXR ligands influence heterodimerization binding affinity between Nur77 LBD and RXR γ LBD. ITC experiments showed no measurable enthalpic component to the Nur77-RXR γ LBD interaction in the absence of ligand at 25 °C, a weak enthalpic component to binding at 15 °C, and a more notable enthalpic component at 5 °C (Figure 3—figure supplement 5 [↗](#)). We could faithfully determine a binding affinity for the apo/ligand-free Nur77-RXR γ LBD heterodimer at 5 °C ($K_d = 9 \mu\text{M}$); by comparison, ITC-determined Nurr1-RXR α LBD heterodimer affinity is $\sim 4 \mu\text{M}$ at 25°C (Yu et al., 2023 [↗](#)). However, addition of select RXR ligands resulted in a muted ITC enthalpy profile that could not be faithfully fitted (Figure 3—figure supplement 6 [↗](#)), indicating RXR ligand binding generally weakens Nur77-RXR γ LBD heterodimer interaction. Although these ITC results are qualitative, they are consistent with our NMR findings indicating that RXR ligand binding promotes dissociation of the Nur77-RXR γ LBD heterodimer to various degrees.

Finally, size exclusion chromatography (SEC) studies revealed that RXR ligands that activate Nurr1-RXR α transcription release a population of monomeric Nurr1 LBD monomer from the Nurr1-RXR α LBD heterodimer (Yu et al., 2023 [↗](#)). We therefore performed SEC to assess how the RXR ligand set influences the oligomeric state of the Nur77-RXR γ LBD heterodimer (Figure 3d [↗](#)). Pharmacological RXR agonists that activate Nur77-RXR γ transcription dissociated a Nur77 LBD monomer species from the Nur77-RXR γ LBD heterodimer in the SEC profiles. Notably, the Nur77 LBD monomer species freed from the Nur77-RXR γ LBD heterodimer appears left shifted compared to the elution profile of Nur77 LBD alone. In our previous Nurr1-RXR α study, the RXR ligand-induced freed Nurr1 LBD monomer species eluted at a volume similar to the profile of Nurr1 LBD alone. Taken together, this indicates the Nur77 LBD monomer population freed by RXR ligand binding likely exchanges with the Nur77-RXR γ LBD heterodimer more rapidly than Nurr1-RXR α , which is consistent with the NMR data that shows a mixture of RXR ligand-induced exchange regimes for Nur77-RXR γ but predominantly slow exchange for Nurr1-RXR α . Interestingly, BRF110 and HX600, two compounds reported as selective agonists of Nurr1-RXR α , also dissociated the Nur77-RXR γ LBD heterodimer and freed a population of Nur77 LBD monomer. Finally, most of the RXR antagonists did not influence the Nur77-RXR γ SEC profile, except for two compounds. UVI3003 displayed a complex SEC profile similar to what we previously observed for Nurr1-RXR α , whereas PA452, which did not dissociate Nurr1-RXR α , induced a population of freed Nur77 LBD monomer, albeit to a lesser degree than RXR agonists.

Correlation analysis reveals the importance of a functionally diverse RXR ligand to illuminate the LBD heterodimer dissociation mechanism

Our findings indicate the RXR ligand set may affect Nur77-RXR γ transcription through classical pharmacological activation and LBD heterodimer dissociation. This prompted us to reconsider our previous observations with Nurr1-RXR α , where the lack of correlation between Nurr1-RXR α transcription and pharmacological RXR α agonism illuminated the LBD heterodimer dissociation mechanism (Yu et al., 2023 [↗](#)). Close inspection of the correlation plots in our previously Nurr1-RXR α study revealed two compounds that stand out in the ligand profiling data: the Nurr1-RXR α selective agonists BRF110 and HX600. When these two Nurr1-RXR α selective agonists are excluded from the correlation analysis, a significant correlation emerges between Nurr1-RXR α transcription and features of pharmacological RXR agonism including RXR α transcription (Figure 4a [↗](#)) and RXR α LBD coactivation interaction in the TR-FRET biochemical assay data (Figure 4b [↗](#)).

We previously used principal component analysis (PCA) to reveal in an unbiased way that Nurr1-RXR α transcriptional activation by the RXR ligand set is not correlated with pharmacological RXR α agonism (RXR α LBD TR-FRET and RXR α transcription) but instead correlated to features of LBD heterodimer dissociation (NMR and ITC data) (Yu et al., 2023 [↗](#)). We re-performed PCA using our previously published Nurr1-RXR α experimental data (Figure 4c [↗](#)) and found that excluding for BRF110 and HX600 resulted in a correlation between Nurr1-RXR α transcriptional activation and both pharmacological RXR α agonism and LBD heterodimer dissociation (Figure 4d [↗](#)). The RXR ligand set used here does not show any selective Nur77-RXR γ

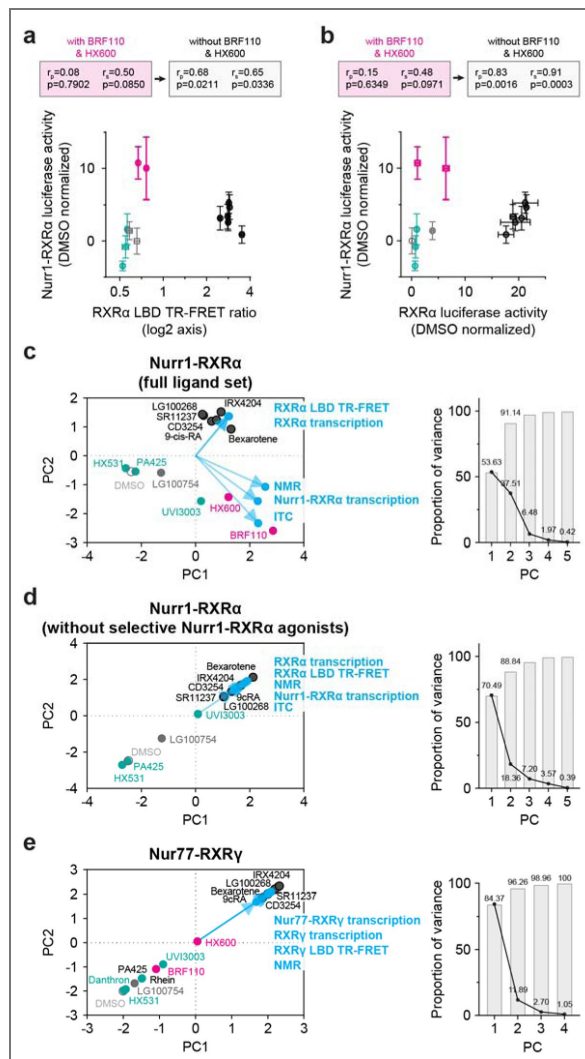


Figure 4. Reanalysis of published Nurr1-RXRα correlation data excluding Nurr1-RXRα selective agonists

(a,b) Correlation plots of our previously reported (a) Nurr1-RXRα cellular transcription data vs. RXRα LBD TR-FRET data and (b) Nurr1-RXRα cellular transcription data vs. RXRα LBD cellular transcription data. Pearson (r_p) and Spearman (r_s) correlation coefficients calculated with or without the two Nurr1-RXRα specific agonists, BRF110 and HX600 (pink data points). (c-e) Principal component analysis (PCA) 2D biplots (left) and proportion of variance plots (right) for our previously published Nurr1-RXRα ligand profiling data (c) including or (d) excluding the Nurr1-RXRα selective agonists BRF110 and HX600; and the (e) Nur77-RXRγ ligand profiling data from this study. Biplots contain the loadings (data types; blue text and blue circles) and ligand-specific PC scores of the first two PCs. Data and RXR ligand label text is colored according to RXR ligand activity as grouped in Figure 2—figure supplement 1 [↗](#).

transcriptional features, which would be apparent if a ligand activated Nur77-RXR γ transcription but showed no change in RXR α homodimer transcription (Figure 2). Consistent with the lack of selective Nur77-RXR γ agonists in the RXR ligand set, PCA analysis revealed that Nur77-RXR α transcriptional activation is correlated to features of pharmacological RXR α agonism and LBD heterodimer dissociation (Figure 4e).

Discussion

Nur77 and its closely related NR4A subfamily member Nur77 are constitutively active orphan nuclear receptors (McMorrow and Murphy, 2011; Zhao and Bruemmer, 2010). We recently revealed that Nur77-RXR α activation occurs through an LBD heterodimer dissociation mechanism, which is distinct from the classical pharmacological NR activation mechanism. This led us to explore the mechanism by which RXR ligands regulate Nur77 activation through targeting its heterodimer partner, RXR γ .

Our data here, together with previous studies (Aarnisalo et al., 2002; Forman et al., 1995; Yu et al., 2023), show that RXRs repress transcription mediated by Nur77 and Nur77. In our RXR γ truncation studies, the RXR γ LBD appeared important for repression of Nur77 activity, as the construct lacking the LBD (RXR γ Δ LBD) did not repress Nur77-mediated transcription. This is consistent with our published observations for Nur77-RXR α , in which the RXR α LBD was sufficient to repress Nur77-mediated transcription and contributed to the discovery of the RXR ligand-induced LBD heterodimer dissociation mechanism (Yu et al., 2023). However, unlike Nur77-RXR α , the RXR γ LBD was not sufficient for repression of Nur77-mediated transcription. One possible explanation is suggested by the behavior of the RXR γ Δ NTD construct, which showed higher Nur77-mediated transcription than wild-type RXR γ , whereas our previous study demonstrated the RXR α Δ NTD construct showed reduced repression Nur77-mediated transcription (Yu et al., 2023). Together, these observations are consistent with a model in which the RXR γ LBD contributes to functionally relevant interaction with Nur77, while the RXR γ NTD contributes to repression. However, the present data do not definitively establish the RXR γ LBD as the only protein-protein interaction interface for Nur77, nor do they resolve the mechanism by which the RXR γ NTD influences transcription. Notably, the NTDs of RXR β and RXR γ are capable of undergoing liquid-liquid phase separation *in vitro* (Softys and Ozyhar, 2023, 2020), and RXR α forms biomolecular condensates in cells with another permissive heterodimer partner, PPAR γ , to promote gene expression (Li et al., 2022). Thus, one possibility is that removal of the RXR γ NTD alters condensate formation in cells or changes the physical properties of RXR γ -containing condensates, thereby influencing the repressive function observed for wild-type RXR γ on Nur77-mediated transcription. In addition, Nur77 has been observed in cellular condensates related to its functions outside of the nucleus (Peng et al., 2021). Future studies are needed to determine how the NTDs of these NRs influence their localization and function within biomolecular condensates and recruitment of other coregulator proteins in cells.

Our previous discovery of the RXR ligand-induced Nur77-RXR α LBD heterodimer dissociation mechanism was possible because we were able to assemble a functionally diverse RXR ligand set that included classical pharmacological agonists and antagonists of RXR homodimers as well as two selective Nur77-RXR α agonists. BRF110 and HX600 compounds are unique in that they activate Nur77-RXR α heterodimer transcription, but display antagonist activity towards RXR α homodimers. Unfortunately, no selective Nur77-RXR γ agonists that activate Nur77-RXR γ transcription but antagonize RXR γ homodimer transcription have been reported. BRF110 was previously shown to activate transcription of Nur77-RXR α heterodimer but not other NR-RXR heterodimers including Nur77-RXR γ (Spathis et al., 2017), which is consistent with our data. On the other hand, data here and in our Nur77-RXR α publication (Yu et al., 2023) shows that HX600 functions as an agonist of Nur77-RXR α and Nur77-RXR γ with an interesting profile for RXR α and RXR γ homodimers. HX600 partially activates RXR α and RXR γ homodimer transcription but does not enhance PGC1 α coactivator peptide interaction to the RXR α LBD or RXR γ LBD in the TR-FRET assay. Although PGC1 α is an important coactivator in RXR-mediated transcription (Delerive et al., 2002), RXRs interact with other coactivators in a ligand-dependent manner including thyroid

hormone receptor-associated proteins (TRAP) and steroid receptor coactivators (SRCs) (Johnson and O'Malley, 2012; Kurcinski and Kolinski, 2010; Malik et al., 2004). It is possible that HX600 modulates interactions with a different set of coactivator proteins rather than PGC1 α . Taken together, our findings suggest that BRF110 and HX600 may alter the conformation or interaction properties of RXR α and RXR γ LBDs in a manner that disrupts their interaction with Nur1 and Nur77 LBDs, respectively. Additional studies into the mechanism of action of BRF110 and HX600 are warranted and may lead to a better understanding of how to elicit selective Nur77-RXR γ activation.

In summary, our published findings (Yu et al., 2023) and new data presented here indicate that transcriptional activation of NR4A-RXR heterodimers by RXR ligands is explained in the case of Nur1-RXR α , and is consistent in the case of Nur77-RXR γ , with a non-classical mechanism involving LBD heterodimer dissociation. Our findings enhance understanding of ligand-dependent NR functions, in particular regarding NR-RXR heterodimers, adding to and reaffirming the complexity and diversity of ligand-dependent NR regulatory mechanisms.

Materials and methods

Materials and reagents

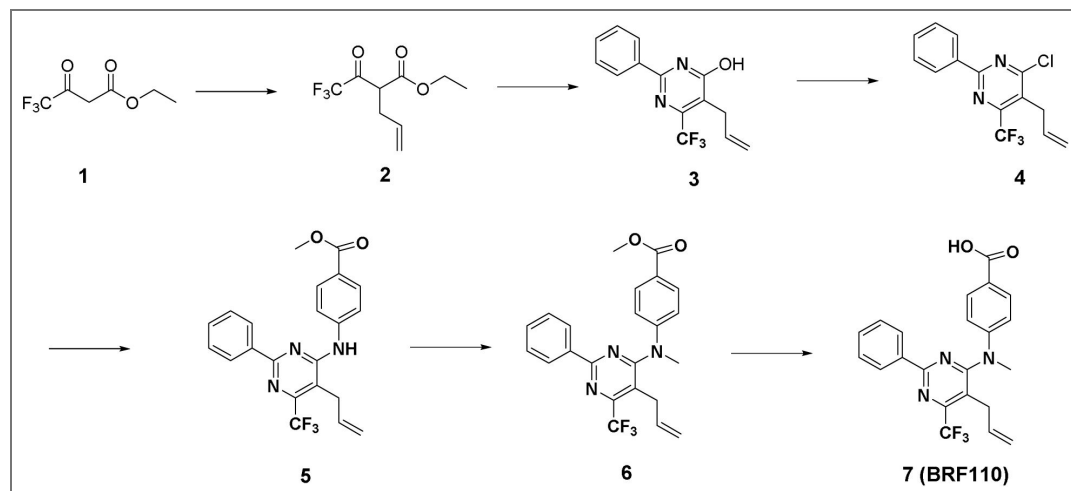
All ligands except BRF110 were obtained from commercial vendors including Cayman Chemicals, Sigma, Axon Medchem, MedChemExpress, or Tocris Bioscience: HX600 (CAS 172705-89-4), 9cRA (CAS 5300-03-8), Bexarotene (CAS 153559-49-0), LG100268 (CAS 153559-76-3), CD3254 (CAS 196961-43-0), SR11237 (CAS 146670-40-8), UVI3003 (CAS 847239-17-2), LG100754 (CAS 180713-37-5), IRX4204 (CAS 220619-73-8), Rhein (CAS 478-43-3), HX531 (CAS 188844-34-0), Danthron (CAS 117-10-2), and PA452 (CAS 457657-34-0). FITC-labeled LXXLL-containing peptide derived from human PGC1 α (137-155; EAEEPSLLKKLLAPANTQ) was synthesized by LifeTein with an N-terminal FITC label and an amidated C-terminus for stability. Bacterial expression plasmids included human Nur77 (NR4A1) LBD (residues 356–598) inserted into a pET45b(+) plasmid (Novagen) as a 3C-cleavable N-terminal hexahistidine (6xHis)-tag fusion protein; and human RXR γ (NR2B3) LBD (residues 226–463) inserted into a pET45b(+) plasmid (Novagen) as a TEV-cleavable N-terminal hexahistidine (6xHis)-tag fusion protein. Luciferase reporter plasmids included a 3xNBRE-luciferase plasmid containing three copies of the NGFI-B response element corresponding to the monomeric binding site for Nur77 (de Vera et al., 2016; Murphy et al., 1996); and a 3xDR1-luciferase containing three copies of the optimal direct repeat 1 (DR1) binding site for RXR γ homodimers (Hughes et al., 2014; Osz et al., 2015; Subauste et al., 1994). Mammalian expression plasmids included full-length human Nur77 (residues 1–598) and full-length human RXR γ (residues 1–463) subcloned into a pcDNA3.1 vector. To clone the RXR γ Δ LBD construct (residues 230–463), site directed mutagenesis and PCR was used to insert a stop codon before the start of the LBD using the full-length RXR α expression plasmid the following primers: forward primer, CTACAGTGGTTAGGAAGACATG; reverse primer, CATGTCTTCCTAACCCTGGTAG. To clone the Δ NTD RXR γ construct (residues 139–463) and RXR γ -hinge-LBD construct (205–463), BamHI was used to cut pcDNA3.1 and subsequently Gibson assembly was used to clone the LBD into the linearized vector using the gBlock sequences in **Supplementary file 1**.

Compound synthesis

BRF110—which was commercially available at the time of our Nur1-RXR α study (Yu et al., 2023) but not commercially available for this current study—was synthesized using a method similar to one that previously described the original synthesis and characterization of BRF110 (Asvos et al., 2025; Spathis et al., 2017), which is briefly described below.

Ethyl 2-(2,2,2-trifluoroacetyl)pent-4-enoate (**2**). Ethyl 4,4,4-trifluoroacetate (**1**, 28.26 g, 153.5 mmol) was added dropwise to a suspension of NaH (6.14 g, 153.5 mmol) in THF (150 mL) at 0 °C. It was stirred for 1 h at 0 °C. The solvent was removed, and anhydrous acetone (70 mL) was added. Subsequently, potassium iodide (2.55 g, 15.36 mmol) was added followed by the dropwise addition of allyl bromide (13 mL, 150 mmol) in anhydrous acetone (40 mL). It was stirred at 60 °C for 48 h.

The solvent was removed, 1 N HCl was added, and it was extracted with DCM (2X). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give the crude product without further purification.



5-allyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4-ol (**3**). Ethyl 2-(2,2,2-trifluoroacetyl)pent-4-enoate (**2**, 33.69 g, 150.28 mmol) was added to a solution of NaOEt (21% w/w in EtOH, 62 ml, 166.07 mmol) and benzamidine hydrochloride (23.54, 150.31 mmol). It was stirred at 100 °C for 16h. The solvent was removed under reduced pressure. It was acidified with 1N HCl and extracted with DCM (2X). It was precipitated from ethanol to yield the title compound.

5-Allyl-4-chloro-2-phenyl-6-(trifluoromethyl)pyrimidine (**4**). 5-allyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4-ol (**3**, 5.4 g, 19.27 mmol) was dissolved in POCl₃ (20 mL). It was stirred at 125 °C for 2h. The solvent was removed under reduced pressure, the residue was dissolved in EtOAc, washed with saturated NaHCO₃ and brine. The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give the crude product, which was purified by flash chromatography on silica gel with EtOAc/Hex (1:10) to obtain the title compound.

Methyl 4-((5-allyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4-yl)amino)benzoate (**5**). 5-Allyl-4-chloro-2-phenyl-6-(trifluoromethyl)pyrimidine (**4**, 259 mg, 0.87 mmol) was dissolved in isopropanol (3 mL). A few drops of concentrated aqueous HCl were added. It was stirred at 140 °C for 48 h. The solvent was removed under reduced pressure, the residue was dissolved in EtOAc, washed with saturated NaHCO₃ and brine. The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give the crude product, which was purified by flash chromatography on silica gel with EtOAc/Hex to obtain the title compound.

Methyl 4-((5-allyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4-yl)(methylamino)benzoate (**6**). Cesium carbonate (86 mg, 0.26 mmol) was added to a solution of methyl 4-((5-allyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4-yl)amino)benzoate (**5**, 81 mg, 0.22 mmol) in DMF (1 mL), followed by the addition of iodomethane (47 mg, 0.33 mmol). It was stirred for 16h at room temperature. It was diluted with EtOAc, washed with saturated NaHCO₃ and brine. The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography to yield the title compound.

4-((5-allyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4-yl)(methyl)amino)benzoic acid (**BRF110**). Methyl 4-((5-allyl-2-phenyl-6-(trifluoromethyl)pyrimidin-4-yl)(methyl)amino)benzoate (**6**, 111 mg, 0.26 mmol) was dissolved in methanol (2 mL), 1 N LiOH (0.57 mL) was added. It was stirred at room temperature overnight. It was diluted with EtOAc, washed with 1 N HCl and brine. The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The residue was purified by silica gel chromatography to yield the title compound. ¹H NMR (400 MHz, d₆-DMSO) δ 12.87 (s, 1H), 8.44-8.38 (m, 2H), 7.94-7.90 (m, 2H), 7.60-7.54 (m, 3H), 7.25-7.21 (m, 2H), 5.65-5.55 (m, 1H), 4.93 (dd, *J* = 1.4, 10.2 Hz, 1H), 4.72 (dd, *J* = 1.6, 17.2 Hz, 1H), 3.59

(s, 3H), 3.00 (d, $J = 5.6$ Hz); ^{13}C NMR (100 MHz, d₆-DMSO) δ 166.70, 165.18, 161.16, 153.31 (q, $J = 32.0$), 149.91, 135.92, 133.70, 131.39, 130.96, 128.79, 127.78, 126.14, 121.34, 119.85, 116.44, 40.75, 29.99; ^{19}F NMR (376.50 MHz, d₆-DMSO) δ -62.89; HRMS (ESI) calculated for $\text{C}_{22}\text{H}_{19}\text{F}_3\text{N}_3\text{O}_2$ [$\text{M} + \text{H}^+$] 414.1429, found 414.1439. LCMS confirmed the purity was >95%.

Cell lines for mammalian cell culture

SK-N-BE(2)-C (#CRL-2268) cells were obtained from and authenticated by ATCC, determined to be mycoplasma free, and cultured according to ATCC guidelines at low passage number (less than 10 passages; typically passages 2–4). Briefly, SK-N-BE(2)-C were grown at 37°C and 5% CO_2 in a media containing 1:1 mixture of EMEM (ATCC) and F12 medium (Gibco) supplemented with 10% fetal bovine serum (Gibco) until 90–95% confluence in T-75 flasks prior to subculture or use.

Protein expression and purification

Proteins were expressed in *Escherichia coli* BL21(DE3) cells using terrific broth (TB) media, autoinduction ZY media, or M9 minimal media (using $^{15}\text{NH}_4\text{Cl}$ and H_2O -based media for ^{15}N -labeled protein; or $^{15}\text{NH}_4\text{Cl}$, ^{13}C -glucose, and D_2O -based media for ^2H , ^{13}C , ^{15}N -labeled protein). For TB expression of Nur77 LBD, cells were grown at 37°C and induced with 1.0 mM isopropyl β -D-thiogalactoside at OD (600 nm) of 1.0, grown for an additional 16 hr at 18°C, and then centrifuged for harvesting. For autoinduction expression of RXR γ LBD, cells were grown for 5 hr at 37°C, then 16 hr at 22°C for RXR γ LBD, then centrifuged for harvesting. For M9 expression of ^{15}N -labeled Nur77 LBD and ^2H , ^{13}C , ^{15}N -labeled Nur77 LBD, cells were grown at 37°C and induced with 1.0 mM isopropyl β -D-thiogalactoside at OD (600 nm) of 0.6, grown for an additional 16 hr at 18°C, and then centrifuged for harvesting. Cell pellets were lysed using sonication and proteins were purified using Ni-NTA affinity chromatography and gel filtration/SEC. The purified proteins were verified by SDS-PAGE, then stored in a buffer consisting of 20 mM potassium phosphate (pH 7.4), 50 mM potassium chloride, and 0.5 mM EDTA. All studies that used RXR γ LBD protein used pooled SEC fractions for the apo-homodimeric form.

Transcriptional reporter assays

SK-N-BE(2)-C cells were seeded in 9.5 cm² cell culture well (Corning) at 0.5 million for transfection using Lipofectamine 2000 (Thermo Fisher Scientific) and Opti-MEM with an empty vector control or full-length Nur77 expression plasmid (2 μg), with or without full-length or different RXR γ truncation constructs (4 μg), and 3xNBRE-luc (6 μg). After incubation for 16–20 hr, cells were transferred to white 384-well cell culture plates (Thermo Fisher Scientific) at 0.5 million cells/mL in 20 μL total volume/well. After a 4 hr incubation, cells were treated with 20 μL of vehicle control (DMSO) or 1 μM ligand. After a final 16–20 hr incubation, cells were harvested with 20 μL Britelite Plus (PerkinElmer), and luminescence was measured on a BioTek Synergy Neo multimode plate reader. The luminescence readouts were normalized to cells transfected with the empty vector (truncation construct assay) or cells treated with DMSO (assays with or without ligand treatment). Data were plotted using GraphPad Prism. In [Figure 1](#), statistical testing was performed and p-values were calculated in GraphPad Prism using the Brown-Forsythe and Welch multiple comparisons test (does not assume equal s.d. values among compared conditions) relative to full-length Nur77 condition. Data are representative of two or more independent experiments ($n=6$ or 9 biological replicates, as indicated in figure legends).

Time-resolved fluorescence resonance energy transfer (TR-FRET) assay

Assays were performed in 384-well black plates (Greiner) using 22.5 μL final well volume. Each well contained 4 nM 6xHis-RXR γ LBD, 1 nM LanthaScreen Elite Tb-anti-His antibody (Thermo Fisher #PV5895), and 400 nM FITC-labeled PGC1 α peptide in a buffer containing 20 mM potassium phosphate (pH 7.4), 50 mM KCl, 5 mM TCEP, and 0.005% Tween 20. Ligand stocks were prepared via serial dilution in DMSO and added to wells (10 μM final concentration) in triplicate. The plates were read using BioTek Synergy Neo multimode plate reader after incubation at 4°C for at least 2

hr. The Tb donor was excited at 340 nm, and its emission was measured at 495 nm; the emission of the acceptor FITC was measured at 520 nm. Data were plotted using GraphPad Prism as TR-FRET ratio (measurement at 520 nm/measurement at 495 nm) at a fixed ligand concentration (2–4 μ M depending on the starting compound stock concentration). Data are representative of two or more independent experiments (n=3 biological replicates).

Size exclusion chromatography with multi-angle light scattering (SEC-MALS)

Gel filtration purified Nur1 LBD, Nur77 LBD, RXR α LBD, and RXR γ LBD were analyzed by SEC-MALS using an HPLC system (Agilent Technologies 1260 Infinity) connected to a MALS system (Wyatt DAWN HELEOS II Ambient with Optilab TrEX HC differential refractive index detector). The SEC-MALS system was calibrated with bovine serum albumin (BSA) before the measurement. Protein was loaded onto pre-equilibrated SEC analytical column (WTC-050S5, Wyatt) with buffer containing 40 mM potassium phosphate (pH 7.4), 200 mM KCl. An aliquot of 100 μ L of each protein at concentration of 5 mg/ml was injected with a flow rate of 0.5 ml/min. All experiments were performed at room temperature (25 °C). Data collection and SEC-MALS analysis were performed with ASTRA 8.0.0.19 (64-bit, Wyatt Technology).

Isothermal titration calorimetry

Experiments were performed using a TA instrument affinity ITC. All experiments were solvent-matched and contained 0.25% DMSO (ligand vehicle) final concentration. RXR γ LBD was preincubated with two equivalent of vehicle (DMSO) or ligand at 5°C. ITC experiments were performed in duplicate by titrating Nur77 LBD to RXR γ LBD in a 10:1 ratio (500 μ M Nur77 LBD to and 50 μ M RXR γ LBD). NITPIC software (Keller et al., 2012) was used to calculate ITC data baselines, integrate curves, prepare experimental data for fitting in SEDPHAT (Brautigam et al., 2016) to obtain binding affinities and thermodynamic parameter measurements, in the case of studies performed without RXR ligand, using a homodimer competition model ($A + B + C \rightleftharpoons AB + C \rightleftharpoons AC + B$; competing B and C for A, where A and C is RXR γ LBD monomer and B is Nur77 LBD monomer. Addition of RXR ligands resulted in an apparent weakening of the Nur77 LBD interaction with RXR γ LBD, which was apparent in the significantly changed and muted enthalpic response that could not be faithfully fit to obtain binding affinities. Final figures were exported using GUSI (Brautigam, 2015).

NMR spectroscopy

For NMR chemical shift assignment, ^2H , ^{13}C , ^{15}N -labeled Nur77 LBD was used to collect two-dimensional (2D) and three-dimensional (3D) backbone assignment data using TROSY-based HSQC, HNCO, HN(CA)CO, HNCA, HN(CO)CA, HN(CA)CB, HN(COCA)CB, and CC(CO)NH-TOCSY experiments performed at 298 K on a Bruker 700 MHz NMR instrument equipped with a QCI cryoprobe using Bruker Topspin (version 2). NMR assignments were performed using RunAbout in NMRViewJ (Johnson, 2018, 2004). For NMR structural footprinting studies, two-dimensional (2D) [^1H , ^{15}N]-TROSY-HSQC NMR experiments were performed at 298 K on a Bruker 900 MHz NMR instrument equipped with a TCI cryoprobe. Samples were prepared in a buffer containing 20 mM potassium phosphate (pH 7.4), 50 mM potassium chloride, 0.5 mM EDTA, and 5% dimethyl sulfoxide- d_6 (DMSO- d_6). Data were collected using Bruker Topspin (version 2) using 200 μ M ^{15}N -labeled Nur77 LBD with or without two molar equivalents of unlabeled RXR γ LBD in the absence or presence of RXR ligands. Data were processed and analyzed using NMRFX (version 11.4.x) (Norris et al., 2016) using unpublished NMR chemical shift assignments for the Nur77 LBD obtained from our lab. Relative Nur77 LBD monomer populations were estimated by the relative peak intensities of the monomeric (I_{monomer}) and heterodimer ($I_{\text{heterodimer}}$) species using the following equation: $I_{\text{monomer_population}} = I_{\text{monomer}} / (I_{\text{monomer}} + I_{\text{heterodimer}})$.

Analytical SEC

To prepare Nur77-RXR γ LBD heterodimer for analytical SEC analysis, purified Nur77 LBD and RXR γ LBD (each 700 μ M in 2.5 mL) were incubated together at 4°C in their storage buffer containing 20 mM potassium phosphate (pH 7.4), 50 mM potassium chloride, and 0.5 mM EDTA for 16 hr, injected (5 mL total) into HiLoad 16/600 Superdex 75 pg (Cytiva) connected to an AKTA FPLC system (GE Healthcare Life Science). Gel filtration was performed using the same buffer to purify Nur77-RXR γ LBD heterodimer for analysis, collecting 2 mL fraction to isolate heterodimer population into 15 samples of 0.5 mL (at a protein concentration of 300 μ M) for analytical gel filtration. Additionally, purified Nur77 LBD and RXR γ LBD pooled gel filtration fractions consisting of the homodimeric species were also used. Samples containing RXR γ LBD (homodimer or heterodimer with Nur77 LBD) were incubated with two molar equivalents of each ligand for 16 hr at 4°C before injection onto Superdex 75 Increase 10/300 GL (Cytiva) connected to the same AKTA FPLC system. The UV chromatograms were exported in CSV format and plotted.

AlphaFold3 model of Nur77-RXR γ

A structural model of the human Nur77-RXR γ LBD heterodimer was generated using AlphaFold3 webserver (<https://alphafoldserver.com>) (Abramson et al., 2024) using the same sequences from our bacterial protein expression constructs. PyMOL (version 3) was used for structural visualization and plotting.

Correlation and other statistical analyses

Correlation plots were performed using GraphPad Prism to calculate Pearson (r_p) and Spearman (r_s) correlation coefficients and two-tailed p-values between two experimental measurements per plot. Statistical testing of control to variable conditions was performed using one-way ANOVA testing as detailed in the figure legends. PCA was performed in GraphPad Prism; all experimentally determined data were used as variables (method = standardize, PCs selected based on parallel analysis at the 95% percentile level with 1000 simulations). p-value statistical significance shorthand, where present, conforms to GraphPad Prism standards: n.s., $p > 0.05$; *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; and ****, $p \leq 0.0001$.

Supplementary figures

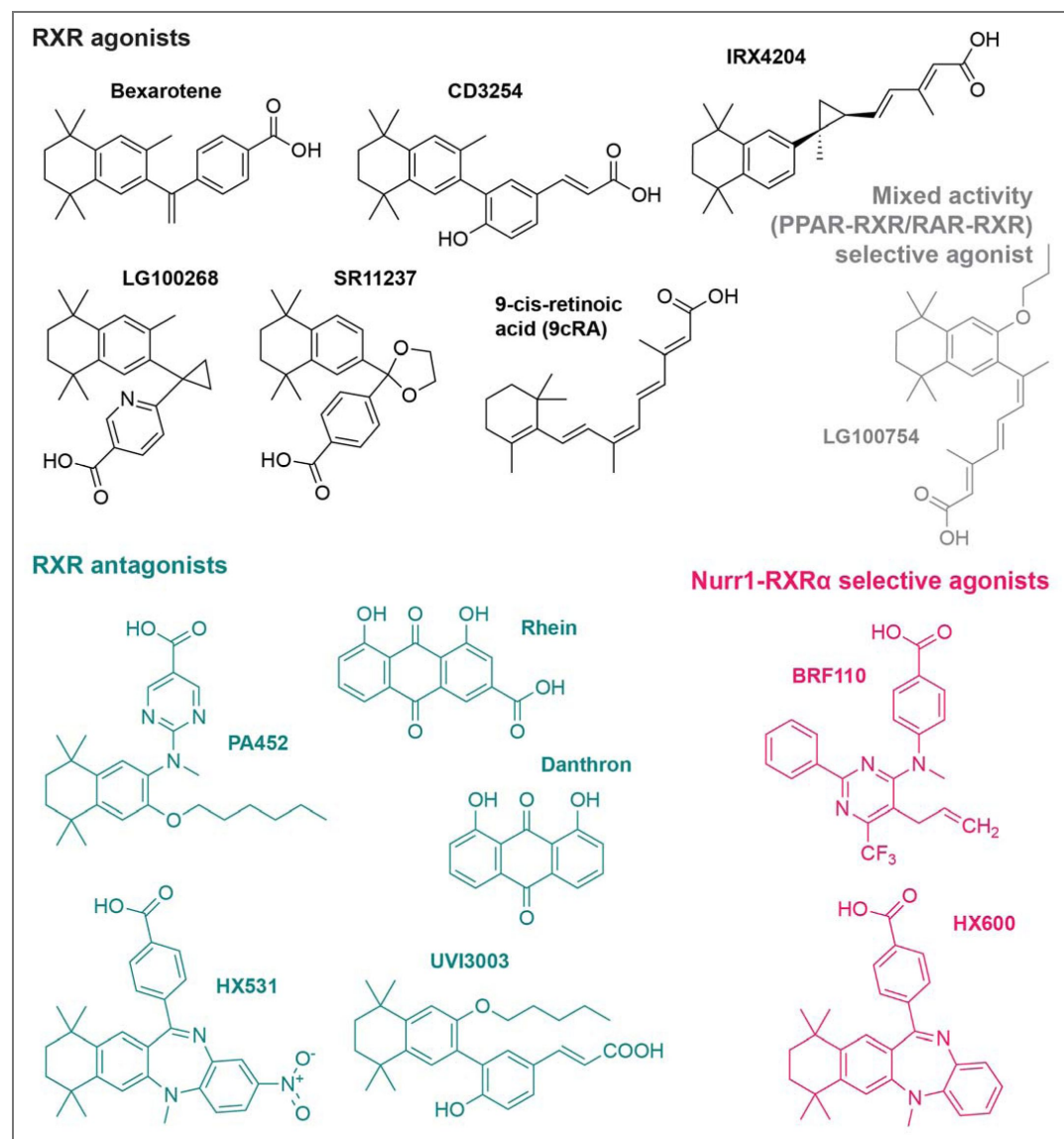


Figure 2 figure supplement 1. RXR ligand set used in this study. Figure was previously published in doi: 10.7554/eLife.85039 as Figure 2; this article was distributed under the terms of the CC BY 4.0 (Attribution 4.0 International) license, which permits unrestricted use and redistribution provided that the original author and source are credited.

Figure 3 figure supplement 1. Multiangle light scattering (MALS) analysis of the ligand-binding domains (LBDs) of Nurrl, Nur77, RXR α , and RXR γ .

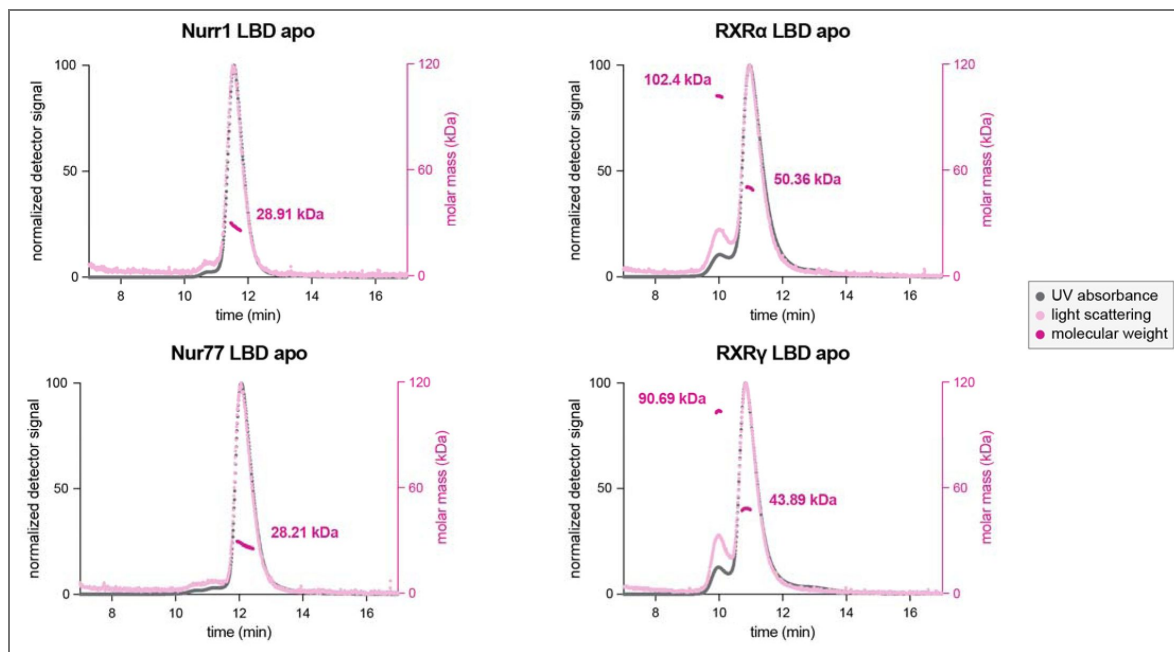


Figure 3 figure supplement 2. Overlay of 2D [^1H , ^{15}N]-TROSY HSQC data of ^{15}N -labeled Nur77 LBD heterodimerized with unlabeled RXR γ LBD.

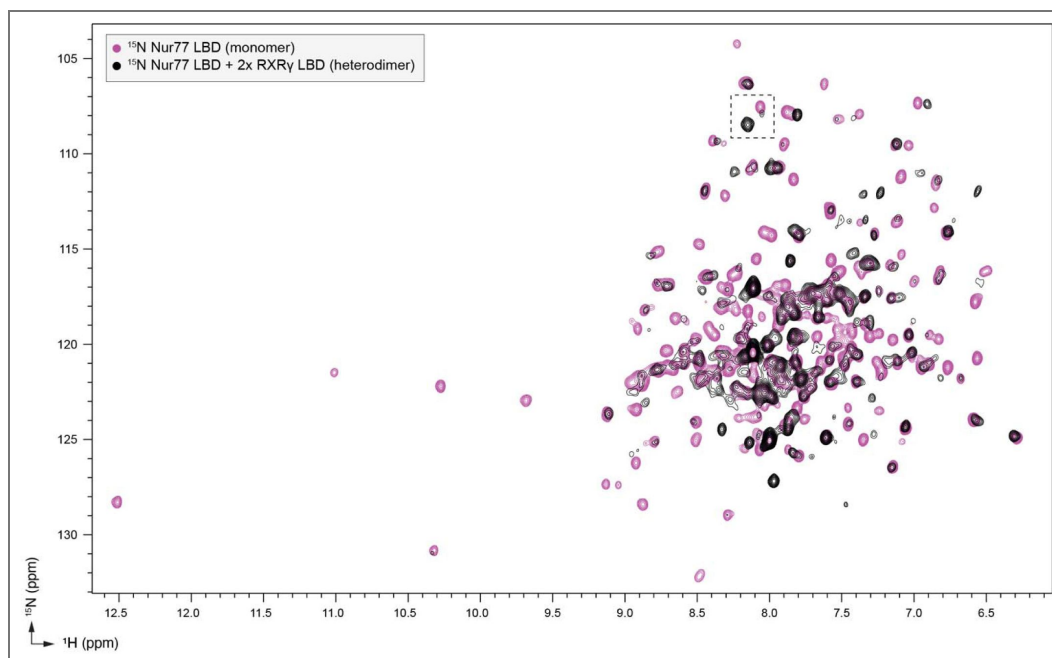


Figure 3 figure supplement 3.

(a) 2D [^1H , ^{15}N]-TROSY HSQC data of ^{15}N -labeled Nur77 LBD heterodimerized with unlabeled RXR γ LBD in the presence of RXR ligands focused on the NMR peak of G376. The upper left shows an overlay of two spectra corresponding to ^{15}N -labeled Nur77 LBD monomer (200 μM ; purple) and ^{15}N -labeled Nur77 LBD + unlabeled RXR γ LBD heterodimer (1:2 molar ratio; black) to demonstrate the shift of the G376 peak between Nur77 LBD monomer (m) and heterodimer (hd) forms; solid green and dotted blue arrows denote the complex chemical shift perturbation pattern. RXR ligand label text is colored according to RXR ligand activity as grouped in Figure 2—figure supplement 1. (b) Peak intensity estimated ligand-dependent Nur77 LBD monomer populations from G544 and G376 in the 2D [^1H , ^{15}N]-TROSY HSQC data. Data are colored according to RXR ligand activity as grouped in Figure 2—figure supplement 1. See Figure 3—source data 1 for data plotted.

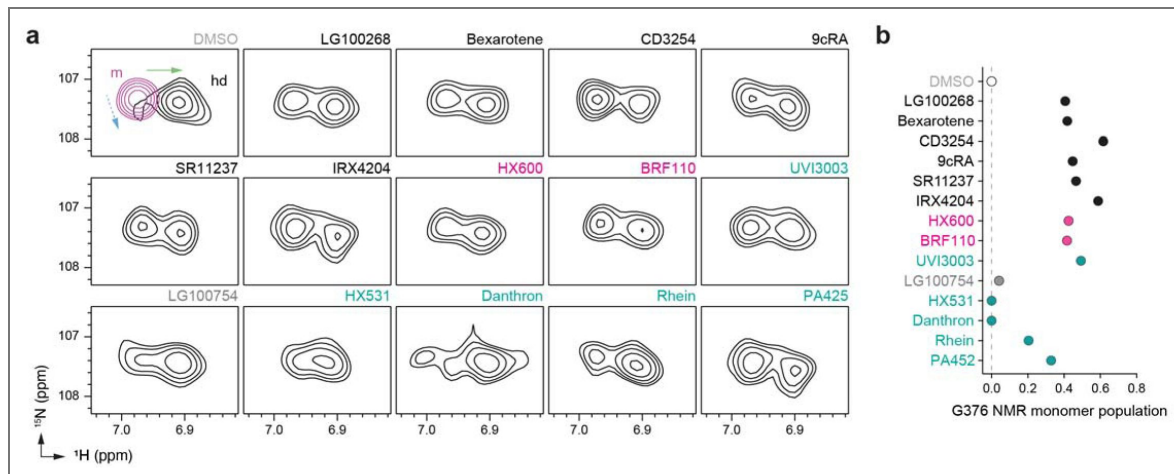


Figure 3 figure supplement 4. AlphaFold3 structural model of the Nur77-RXR γ LBD heterodimer used to highlight the locations of G376 and G544 that were analyzed in the NMR analysis.

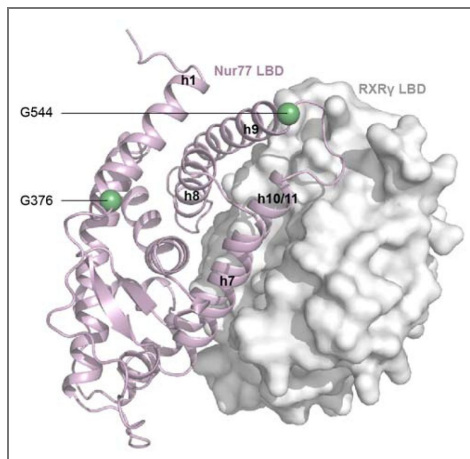


Figure 3 figure supplement 5. Isothermal titration calorimetry (ITC) analysis of Nur77 LBD titrated into RXR γ LBD at the indicated temperatures.

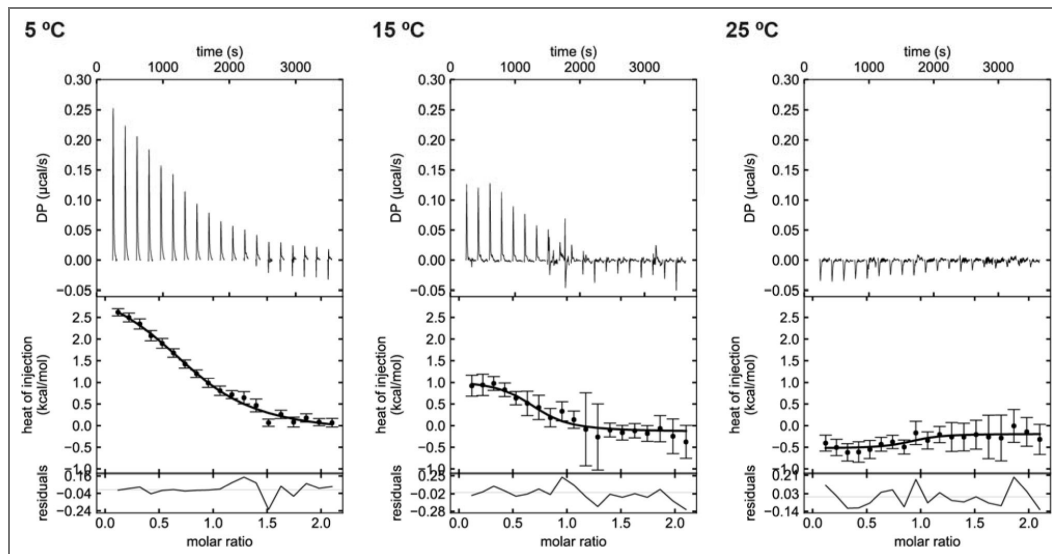
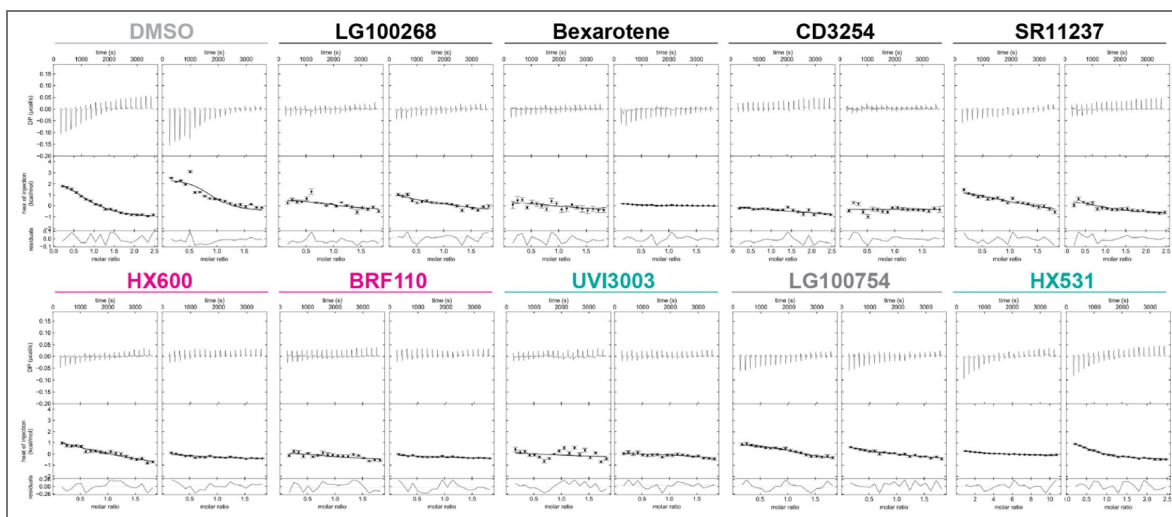


Figure 3 figure supplement 6. Isothermal titration calorimetry (ITC) analysis of Nur77 LBD titrated into RXR γ LBD performed at 5 °C in the presence of the indicated RXR ligands or DMSO (vehicle control).



Data availability

NMR chemical shift assignments have been deposited in the Biological Magnetic Resonance Bank (BMRB) under accession code 52973. AlphaFold3 model of the Nur77-RXR γ LBD is available as **Supplementary file 2**. All other data generated or analyzed during this study are included in the manuscript as Source Data files or, in the case of Figure 4a-d [was previously published](#) (doi: 10.7554/eLife.85039).

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Additional information

Author contributions

X.Y. performed the research. Y.H. and T.M.K. synthesized BRF110. D.J.K. supervised the research. X.Y. and D.J.K. conceived and designed the research, analyzed data, and wrote the manuscript.

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Additional files

[Source Data](#) [↗](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

This foundational study builds on prior work from this group to reveal the complexities underlying ligand-dependent RXR γ -Nur77 heterodimer formation, offering a compelling re-evaluation of their earlier conclusions. The Authors examine how a library of RXR ligands influences the biophysical, structural, and functional properties of Nur77. They find that although the Nur77-RXR γ heterodimer shares notable functional similarities with the Nurrr1-RXR α complex, it also exhibits unique features - notably, both dimer dissociation and classical agonist-driven activities. This work advances our understanding of the nuanced behaviors of nuclear receptor heterodimers, which have important implications for health and disease.

Strengths:

- (1) Builds on previous work by providing a comprehensive analysis that examines whether Nur77-RXR γ heterodimer formation parallels that of the Nurrr1-RXR α complex.
- (2) Systematic evaluation of a library of RXR ligands provides a broad survey of functional outputs.
- (3) Careful reanalysis of previous work sheds new light on how NR4A heterodimers function.

Weaknesses:

- (1) Some conclusions appear overstated or are not well substantiated by the work presented. It's unclear how the data support a non-classical mode of agonism, for example, based on the data shown.
- (2) Some assays have relatively few replicates, with only two in some cases.

Comments on revisions:

I'm satisfied with the revised version.

<https://doi.org/10.7554/eLife.106861.2.sa2>

Reviewer #2 (Public review):

Summary:

This study explores the mechanisms by which binding of the nuclear receptor RXR γ regulates its heterodimeric partner Nur77. Previously, this group made the interesting discovery that ligand-dependent activation of RXR γ bound to a related partner, Nurrr1, does not occur through a classical pharmacological mechanism but through agonist-dependent dissociation of the complex through disruption of their ligand binding domain (LBD) interactions. Here, they revisit this paradigm with Nur77. In contrast to Nurrr1, the authors do not have the reagents to clearly support a role for LBD dissociation. Following from the model of partial ligand-dependent dissociation of the LBD heterodimer, the experimental data (NMR, ITC, SEC) are interesting and quite complex.

Strengths:

The authors do a rigorous job of describing the data and providing possible interpretations and caveats. Revisiting the analysis of Nurrr1, they identify the crucial role that selective

Nurr1-RXRg agonists played in supporting the LBD dissociation model; without analogous compounds for the Nur77-RXRg complex, it is difficult to invoke this mechanism. Interestingly, treatment with the Nurr1-RXRg selective agonist HX600 suggests it can induce some LBD dissociation. Therefore, there may be some similarities between regulation of Nurr1 and Nur77 by RXRg.

Weaknesses:

Despite evidence supporting a partial role for RXRg LBD dissociation as a mechanism to activate Nur77, other data demonstrate that a fundamentally different regulatory mechanism likely exists in the Nur77-RXRg complex that involves the RXRg disordered NTD. The decision to describe further study of this as outside the scope of this work is unfortunate, as it closed off an avenue that could have provided fruitful data informing the apparently distinct regulatory mechanisms of the Nur77-RXRg complex. Given the uncertainty in the importance of the partial roles of the pharmacological mechanism, LBD dissociation, and the RXRg NTD, this study may have limited impact on the field.

Comments on revisions:

I'm satisfied with the revision.

<https://doi.org/10.7554/eLife.106861.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This foundational study builds on prior work from this group to reveal the complexities underlying ligand-dependent RXRy-Nur77 heterodimer formation, offering a compelling re-evaluation of their earlier conclusions. The authors examine how a library of RXR ligands influences the biophysical, structural, and functional properties of Nur77. They find that although the Nur77-RXRy heterodimer shares notable functional similarities with the Nurr1-RXRa complex, it also exhibits unique features, notably, both dimer dissociation and classical agonist-driven activities. This work advances our understanding of the nuanced behaviors of nuclear receptor heterodimers, which have important implications for health and disease.

Strengths:

- (1) Builds on previous work by providing a comprehensive analysis that examines whether Nur77-RXRy heterodimer formation parallels that of the Nurr1-RXRa complex.*
- (2) Systematic evaluation of a library of RXR ligands provides a broad survey of functional outputs.*
- (3) Careful reanalysis of previous work sheds new light on how NR4A heterodimers function.*

We thank the reviewer for recognizing our work as foundational. In the nuclear receptor field, current understanding of ligand-regulated nuclear receptor activity is based largely on ligand-dependent coregulator recruitment preferences; for example, agonists enhance coactivator recruitment to activate transcription. Building on our recent study of Nurr1-

RXR α , the present work suggests that activation of the evolutionarily related NR4A-RXR heterodimer Nur77-RXR γ by RXR ligands is also consistent with a non-classical activation mechanism involving heterodimer dissociation.

Weaknesses:

(1) Some conclusions appear overstated or are not well substantiated by the work presented. It's unclear how the data support a non-classical mode of agonism, for example, based on the data shown.

We thank the reviewer for this important point. We did not intend to claim that Nur77-RXR γ activation is explained exclusively by a non-classical mode of agonism. Rather, our interpretation was that the data are consistent with two possible, non-mutually exclusive mechanisms: (1) a classical pharmacological mechanism involving ligand-dependent coregulator recruitment; and (2) a non-classical mechanism involving ligand-binding domain (LBD) heterodimer dissociation, as we previously described for Nur1-RXR α . This differs from our prior eLife study of Nur1-RXR α , in which the data supported the LBD heterodimer dissociation model but not the classical pharmacological model.

In our revised manuscript, we clarify two points that are important for interpreting the Nur77-RXR γ data. First, several experimental limitations of the Nur77-RXR γ studies reduced the extent to which the mechanism could be resolved as rigorously as in our earlier Nur1-RXR α study. Second, and more importantly, the currently available ligand set lacks Nur77-RXR γ -selective agonists. This limits our ability to determine whether LBD heterodimer dissociation is the sole or principal mechanism of activation, or instead one of several contributing mechanisms.

Taken together, these results support LBD heterodimer dissociation as a plausible and experimentally observable component of Nur77-RXR γ activation and, therefore, as a candidate shared activation mechanism for NR4A-RXR heterodimers. At the same time, because the quantitative evidence is less definitive than in the Nur1-RXR α system, we agree that conclusions regarding Nur77-RXR γ should be stated more cautiously. This caution is reflected in both the title of our manuscript (“Towards a unified mechanism...”) and the language used throughout the text.

(2) Some assays have relatively few replicates, with only two in some cases.

We thank the reviewer for their attention to experimental rigor. For some assays, the findings were reproduced in two independent experiments, which we considered sufficient to confirm the presence and reproducibility of the effects observed in those particular assay formats. In the original manuscript, we used a general statement in the figure legends (“representative of two or more independent experiments”) across all assay data. In the revised manuscript, we now specify the number of independent experimental replicates for each assay in the corresponding figure legends to improve transparency.

Reviewer #2 (Public review):

Summary:

This study explores the mechanisms by which binding of the nuclear receptor RXR γ regulates its heterodimeric partner Nur77. Previously, this group made the interesting discovery that ligand-dependent activation of RXR γ bound to a related partner, Nur1, does not occur through a classical pharmacological mechanism but through agonist-dependent dissociation of the complex through disruption of their ligand binding domain (LBD) interactions. Here, they revisit this paradigm with Nur77. In contrast to Nur1, the authors do not have the reagents to clearly support a role for LBD

dissociation. Following the model of partial ligand-dependent dissociation of the LBD heterodimer, the experimental data (NMR, ITC, SEC) are interesting and quite complex.

Strengths:

The authors do a rigorous job of describing the data and providing possible interpretations and caveats. Revisiting the analysis of Nurr1, they identify the crucial role that selective Nurr1-RXRg agonists played in supporting the LBD dissociation model; without analogous compounds for the Nur77-RXRg complex, it is difficult to invoke this mechanism. Interestingly, treatment with the Nurr1-RXRg selective agonist HX600 suggests it can induce some LBD dissociation. Therefore, there may be some similarities between the regulation of Nurr1 and Nur77 by RXRg.

We thank the reviewer for this thoughtful and balanced summary of our work. We appreciate the reviewer's recognition of both our prior findings in the Nurr1-RXR α system and the interesting, but more complex, experimental behavior observed here for Nur77-RXR γ . We agree that the absence of Nur77-RXR γ -selective agonists currently limits how definitively the contribution of LBD dissociation can be resolved, and we have revised the manuscript to make this point more explicit and to further temper our conclusions accordingly.

Weaknesses:

Despite evidence supporting a partial role for RXRg LBD dissociation as a mechanism to activate Nur77, other data demonstrate that a fundamentally different regulatory mechanism likely exists in the Nur77-RXRg complex that involves the RXRg disordered NTD. The decision to describe further study of this as outside the scope of this work is unfortunate, as it closed off an avenue that could have provided fruitful data informing the apparently distinct regulatory mechanisms of the Nur77-RXRg complex. Given the uncertainty in the importance of the partial roles of the pharmacological mechanism, LBD dissociation, and the RXRg NTD, this study may have limited impact on the field.

We thank the reviewer for this thoughtful point. We agree that the RXR γ NTD likely contributes to regulation of Nur77-RXR γ transcription, and that our truncation data suggest that regions outside the LBD can influence transcriptional output. At present, however, the effect of RXR γ NTD truncation is not sufficiently mechanistically resolved to distinguish among several plausible explanations.

For example, the RXR γ NTD has been implicated in phase separation and biomolecular condensate formation in cells (PubMed ID 40392852, 40420113, 33971237, 31881311), and perturbing these properties (via RXR γ NTD truncation) could indirectly affect Nur77-RXR γ transcriptional activity. In addition, NTDs of nuclear receptors can participate in coactivator or corepressor interactions (PubMed ID 24284822), raising the possibility that removal of the RXR γ NTD alters transcription by changing recruitment of regulatory factors rather than by directly informing the LBD-centered mechanism examined here. We will clarify in the revised manuscript that these possibilities remain unresolved and represent important directions for future study.

We also agree that defining how multiple RXR γ domains contribute to Nur77-RXR γ regulation would be valuable for the field. However, the focus of the present study is narrower: to test whether, as in our previous eLife study of Nurr1-RXR α , RXR ligands can influence heterodimer function through effects on LBD-LBD interactions. Because the available data do not yet allow a mechanistic dissection of the RXR γ NTD contribution, we believe that a definitive analysis of this question would require a separate set of experiments beyond the scope of the present work. We have revised the manuscript to better acknowledge this limitation and to frame the conclusions accordingly.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Overall, this is a compelling body of work. Additional summary statements and clearer transitions would be helpful throughout.

Here are some points that should be addressed or at least discussed by the authors:

(1) It is unclear in the luciferase assays whether the truncated proteins are functional or not. Were there Western blots or other assays run to confirm protein concentrations?

We thank the reviewer for this point. We did not perform Western blotting or other assays to confirm equivalent expression levels of the truncated RXR γ constructs, and we agree that this is a limitation of the luciferase assay data. As a result, the transcriptional effects observed with the truncation constructs should be interpreted cautiously.

With that said, the increased transcriptional activity observed upon deletion of the RXR γ NTD/AF-1 region suggests that this region may exert a repressive effect on Nur77-RXR γ transcription. This effect could reflect multiple, non-mutually exclusive mechanisms, including altered phase separation or condensate-related properties of RXR γ , or altered recruitment of transcriptional coregulators through the NTD. Because our truncation strategy does not distinguish among these possibilities, we do not believe these data allow a definitive mechanistic interpretation of the NTD contribution.

We have revised the manuscript to clarify this limitation. We also note that the primary focus of the present study is the role of ligands in modulating Nur77-RXR γ function through LBD-mediated interactions, in direct comparison with our previous Nur77-RXR α study. A more complete mechanistic dissection of how RXR γ domain architecture influences Nur77-RXR γ transcription will require future work.

(2) Why does the Nur77 construct lacking the NTD show increased luciferase activity?

Please see our response above to Reviewer 2's Public Review, which also addresses this point.

(3) A case is made for the Nur77 LBD driving the activity, but it also could be inferred that the DBD is driving based on the data shown in Figure 1.

We thank the reviewer for this point. We agree that the Nur77 DBD is required for binding to NBRE response elements, and we did not intend to suggest otherwise. The experimental approach in Figure 1 was not designed to dissect the relative contributions of Nur77 domains, since Nur77 was tested only in its full-length form. Instead, the purpose of this experiment was to examine how truncation of RXR γ domains affects Nur77-RXR γ transcriptional activity, in direct comparison with our prior eLife study of Nur77-RXR α , where RXR α domain truncations helped define the importance of RXR-LBD-mediated regulation. We will revise the text to clarify that Figure 1 does not distinguish whether Nur77 DBD-dependent DNA binding is necessary, but instead addresses whether the pattern of RXR γ domain dependence is consistent with an LBD-centered mechanism of ligand-regulated heterodimer function.

(4) It is stated that the HX600 coactivator recruitment requires further study. Why wasn't it studied here?

We thank the reviewer for this point. The primary focus of this study was to determine how RXR ligands influence Nur77-RXR γ heterodimer activity, particularly in relation to ligand-dependent effects on heterodimer function. A more detailed analysis of HX600-dependent coactivator recruitment would require a broader mechanistic investigation of RXR α and RXR γ homodimer pharmacology and RXR-specific coregulator interactions, which extends

beyond the central scope of the present manuscript. We agree that this is an important question and view it as a valuable direction for future work.

(5) Figure 3B, the shifts in monomer populations, error bars aren't shown, the biggest shift is from 0.2 to 0.6, is that statistically meaningful?

We thank the reviewer for this point. The reviewer is correct that error bars were not shown for Figure 3B. These NMR measurements were performed once ($n=1$), and therefore the shifts in monomer populations shown in Figure 3B cannot be assessed statistically. Because these studies required substantial NMR instrument time and isotopically labeled protein at high concentration, we were not able to perform experimental replicates for this dataset. We have revised the figure legend to explicitly state that these data were collected from a single experiment and have tempered the corresponding language in the manuscript accordingly.

(6) Some ligands are shown in the figures but don't appear to be discussed in the text (at least that I can find), such as SR11237.

We thank the reviewer for pointing this out. We used a panel of 14 commercially available RXR ligands with different pharmacological properties to probe Nur77-RXR γ function, as in our previous Nurr1-RXR α study. In the text, we emphasized ligands that were most informative for the mechanistic conclusions, rather than discussing every compound individually. SR11237, for example, behaved similarly to the broader group of RXR agonists and was therefore shown as part of the full ligand panel but not specifically highlighted in the text. We will clarify this in the revised manuscript.

(7) There is a sentence in the discussion that says "these observations implicate that although RXR γ LBD provides the protein-protein interaction interface to bind Nur77...." the authors did not show enough data to support this claim. It should be bolstered.

We thank the reviewer for this point. We agree that this statement was stronger than was warranted by the data presented. Our intent was not to claim that the present study definitively establishes the RXR γ LBD as the sole or fully defined protein-protein interaction interface for Nur77 binding. Rather, based on the domain truncation data together with our prior Nurr1-RXR α study, we intended this statement as a working interpretation consistent with an LBD-centered mechanism. In our revised manuscript, we have softened this language to avoid overstating the conclusion and clarified that the current data support, but do not definitively prove, a role for the RXR γ LBD in mediating functionally relevant interaction with Nur77.

Reviewer #2 (Recommendations for the authors):

Even though this study is not able to make definitive claims about the mechanism(s) of activation of Nur77 in the Nur77-RXR γ complex, the work presented here is rigorous and solidly interpreted. Identifying differences between Nurr1 and Nur77 regulation is important, and the work here shows that selective agonists are essential for supporting the non-canonical mechanism they identified before. Although they address potential implications of NTD regulation in the discussion, it feels like a lot of insight into Nur77 regulation is being missed. However, it is clear that addressing this experimentally would require substantially more work. I don't have any specific recommendations. Given current limitations on funding, I think it's fine to focus on the work completed with the acceptance that it likely limits the impact of the work on the field.

We thank the reviewer for this thoughtful and balanced assessment of our work. The goal of this manuscript was to test whether the LBD heterodimer dissociation mechanism that we previously reported for Nurr1-RXR α may represent a conserved feature of NR4A-RXR heterodimers by extending these studies to Nur77-RXR γ . We agree that understanding the role of the RXR γ NTD in Nur77-RXR γ regulation is important and potentially highly

informative. At the same time, resolving that question experimentally would require a distinct and more extensive set of studies beyond the scope of the present work. We have therefore chosen to focus this manuscript on the completed LBD-centered studies, while acknowledging that this narrower scope may limit the broader impact of the work.

Minor points:

(1) Without page and line numbers, it is not easy to point out specific text. On the bottom of page 6 of the document, there are two references to Figure 3a, and the arrows that help illustrate RXRg LBD-dependent CSPs; the second figure callout should describe the blue arrow, I believe.

Thank you, we made this change.

(2) Bottom of page 8, "...revealed two compounds [that] standout..."

Thank you, we made this change.

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